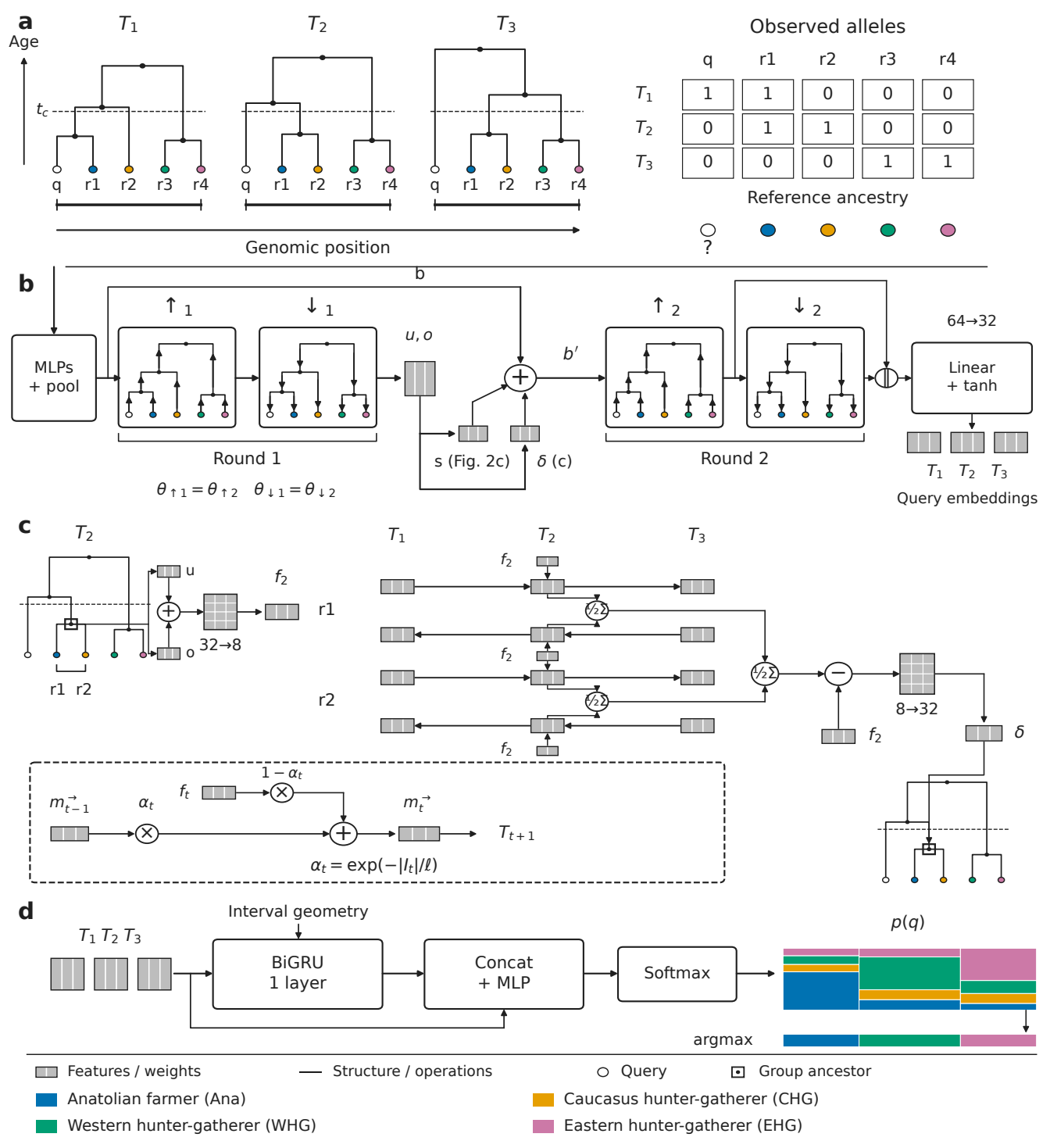

a
Age
t_c
T_1
T_2
T_3
q r1 r2 r3 r4
Genomic position
Observed alleles
q r1 r2 r3 r4
T_1 1 1 0 0 0
T_2 0 1 1 0 0
T_3 0 0 0 1 1
Reference ancestry
?
b
MLPs + pool
↑ 1
↓ 1
u, o
b′
↑ 2
↓ 2
64→32
Linear + tanh
Round 1
$\theta_{\uparrow 1} = \theta_{\uparrow 2}$ $\theta_{\downarrow 1} = \theta_{\downarrow 2}$
s (Fig. 2c)
δ (c)
Round 2
T_1 T_2 T_3
Query embeddings
c
T_2
u
o
f_2
32→8
r1 r2
r1
r2
T_1 T_2 T_3
f_2
½Σ
8→32
δ
$m_{t-1}^{\rightarrow}$
α_t
f_t
$1-\alpha_t$
$m_t^{\rightarrow}$
T_{t+1}
$\alpha_t = \exp(-|l_t|/\ell)$
d
Interval geometry
T_1 T_2 T_3
BiGRU 1 layer
Concat + MLP
Softmax
$p(q)$
argmax
Features / weights
Structure / operations
Query
Group ancestor
Anatolian farmer (Ana)
Caucasus hunter-gatherer (CHG)
Western hunter-gatherer (WHG)
Eastern hunter-gatherer (EHG)